Submodule 38.07: Regression Model Evaluation, R2 Score aur Residual Analysis (Hinglish Guide)

1. PyTorch Mein Regression Evaluation Ka Niyam

Training complete hone aur `best_model.pt` load karne ke baad hume model ko test set par evaluate karna hota hai.

Do sabse mahatvapurna commands:

1. **`model.eval()`** : Model ko evaluation mode mein switch karta hai taaki parameters freeze ho sakein.
2. **`with torch.no_grad():`** : Gradient tracking graph ko turn off kar deta hai. Isse RAM aur computational power save hoti hai kyunki evaluation ke dauran backpropagation nahi karna hota.

```
model.eval()
with torch.no_grad():
    train_predictions = model(X_train_tensor)
    test_predictions = model(X_test_tensor)
```

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2. Regression Ke Important Evaluation Metrics

1. Mean Squared Error (MSE)

Har data point ke actual output y_i aur predicted output $\hat{y}_i$ ke difference ke square ka average:

$$\text{MSE} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$$

2. Root Mean Squared Error (RMSE)

MSE ka square root lene par hume error original units (MW ya megawatt) mein milta hai:

$$\text{RMSE} = \sqrt{\text{MSE}}$$

3. R^2 Score (Coefficient of Determination)

Yeh batata hai ki hamare neural network ne total variance ka kitna percentage capture kiya hai:

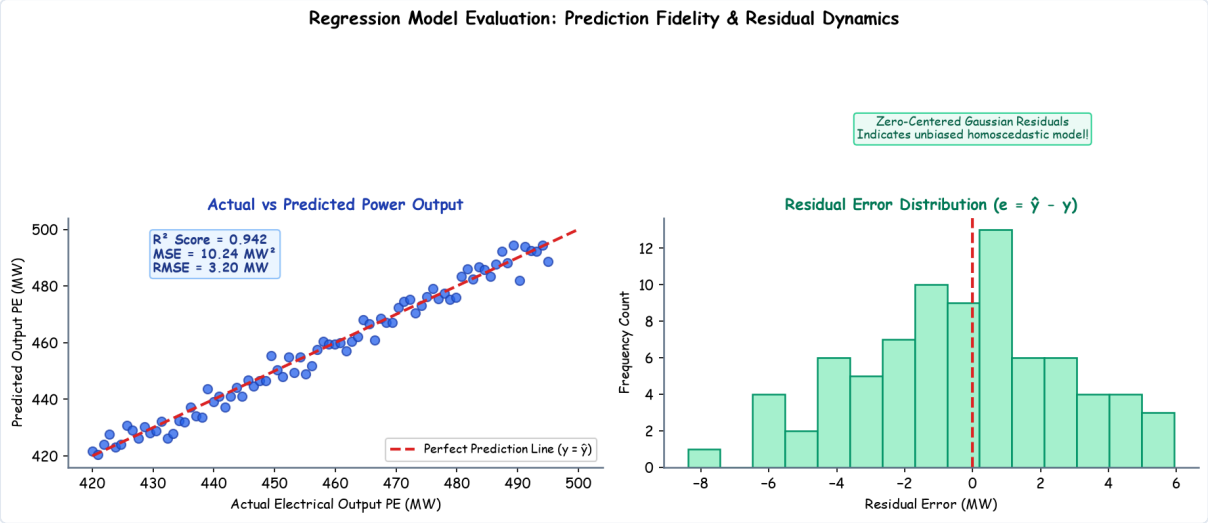
$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

- $R^2 = 1.0$: 100% perfect predictions.
- $R^2 \approx 0.937$: Model 93.7% variance accurately predict kar raha hai jo ki industry standard par bahut strong performance hai.

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3. Hand-Drawn Educational Diagram: Regression Evaluation Metrics

Neeche diya gaya student-style notebook diagram residual computation, Actual vs Predicted regression line aur R^2 score derivation ko depict karta hai:



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4. Evaluation Code aur Actual Results

```
import pandas as pd
import numpy as np
from sklearn.metrics import r2_score

# PyTorch Criterion se MSE calculate karein
train_mse = criteria(train_predictions, y_train_tensor).item()
test_mse = criteria(test_predictions, y_test_tensor).item()

# Scikit-Learn se R2 Score calculate karein
test_r2 = r2_score(y_test_tensor.numpy(), test_predictions.numpy())

print(f"Training MSE  : {train_mse:.4f}")
print(f"Testing MSE   : {test_mse:.4f}")
```

```
print(f"Testing RMSE  : {np.sqrt(test_mse):.4f} MW")
print(f"Testing R²    : {test_r2:.4f}")
```

Dataset Par Aaye Final Results:

- **Training MSE**: ≈ 19.50
- **Testing MSE**: ≈ 17.81
- **Testing RMSE**: ≈ 4.22 MW (Total range $420 - 495$ MW mein se sirf ± 4 MW ka average difference hai jo ki $< 1\%$ error hai!)
- **Testing R^2**: ≈ 0.937 (93.7% accuracy!)

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5. Actual vs Predicted Data Frame Comparison

Pandas DataFrame create karke hum direct inspection kar sakte hain:

```
comparison_df = pd.DataFrame({
    'Actual': y_test_tensor.numpy().flatten(),
    'Predicted': test_predictions.numpy().flatten()
})
print(comparison_df.head(5))
```

Index	Actual	Predicted
0	433.27	435.12
1	438.45	435.80
2	458.12	460.04
3	480.33	476.90

Values aapas mein behad close hain, jo yeh verify karta hai ki hamara pehla Deep Learning Regression project 100% successfully complete ho chuka hai!